# Interpretability of Hybrid Feature Using Graph Neural Networks from Mental Arithmetic Based EEG

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***Abstract*— A high cognitive load could significantly impair problem-solving skills. Electroencephalogram (EEG)-based real-time assessment of mental workload is feasible, and graph neural networks (GNN) can classify brain activity patterns during cognitively demanding tasks with high accuracy. However, previous GNN studies pertaining to mental workload classification lack explainability. This study utilized a state-of-the-art GNN variant with GNNexplainer to find relevant connectivity during mental arithmetic (MA) tasks. In this endeavor, MA EEG recordings were retrieved from an open-access database. The signals were transformed to graph data through the envelope correlation and power spectral density (PSD), and subjected to GNN with hierarchical graph pooling with a structure learning model to classify MA and baseline (BL). The model accuracy was 85.57 ± 6.27 and 96.26 ± 4.14% for the connectivity dataset and the PSD and the connectivity feature, respectively. Among the connections between nodes identified as important by GNNExplainer, two notable edge patterns were found as 1) from the left centro-parietal region to left frontal regions, and 2) the frontoparietal connection. The results indicate 1) the GNN model performance could be improved using the connectivity and PSD feature together, and 2) characteristic patterns of the connectome and PSD could be important for MA classification. The connectivity analysis by the "explainable" GNN model could be beneficial in future brain activity pattern studies.**

***Keywords*— *brain computer interface, mental arithmetic task, electroencephalography, conncetivity, graph neural networks, explainable artificial intelligence***

## I. Intruduction

An increased cognitive load could lead to increased human error and decreased performance in workplaces [1, 2], and temporary attention impairment [3]. Given that an adequate level of attention and cognitive capacity is crucial for problem-solving [4], monitoring and evaluation of cognitive load could be helpful in a wide variety of situations.

Previous studies pertaining to real-time evaluation of mental workload utilized electroencephalogram (EEG) to monitor changes in brain activity patterns during cognitively demanding task [5-7]. In particular, the EEG bands of theta (4-8 Hz) and alpha (8-12 Hz) reflect mental workload status [8, 9]; in motor imagery studies, the difference of left/right hemisphere alpha band powers was found to be associated with the level of attention during the task [10, 11]. Most previous EEG studies pertaining to mental workload focused on finding specific EEG features that are associated with cognitive status; however, the changes in brain activity patterns during tasks could be better assessed by investigating the connectivity across various brain regions.

The network organization of the brain can be understood via graph theory [12]. With graph neural network (GNN), a deep learning method designed to be applied to graphs, recent studies have reported notable classification performance in emotion recognition [13], identifying reading tasks [14], and mental arithmetic tasks [15]. Nonetheless, the existing GNN models do not offer an explanation of brain regions and spectral components that were more relevant in model output.

This study aims to utilize a GNN for the identification of the mental arithmetic (MA) task. In this endeavor, the connectivity and spectral information were extracted from EEG recordings during MA task and subjected to a state-of-the-art GNN model. Further, the GNNExplainer was applied to grant the model an explainability on its output. The methods described in this study are expected to yield high classification performance, and specify the brain regions and EEG bands that were relevant to MA tasks.

## II. Materials and Methods

### A. Dataset Description

An open-access EEG dataset that contains recordings of two mental states, i.e., MA and baseline (BL) were assessed and subjected to analyses [16]. The dataset comprises twenty-nine healthy subjects (14 males, 15 females) with an average age of 28.5 +- 3.7 years. The EEG recordings were acquired from 30 electrodes placed according to the international 10-5 system at a sampling frequency of 200 Hz. The MA experiment consisted of 3 sessions, each comprised of 20 trials. MA and BL trials were presented randomly 10 times in each session. During MA tasks, subjects were instructed to

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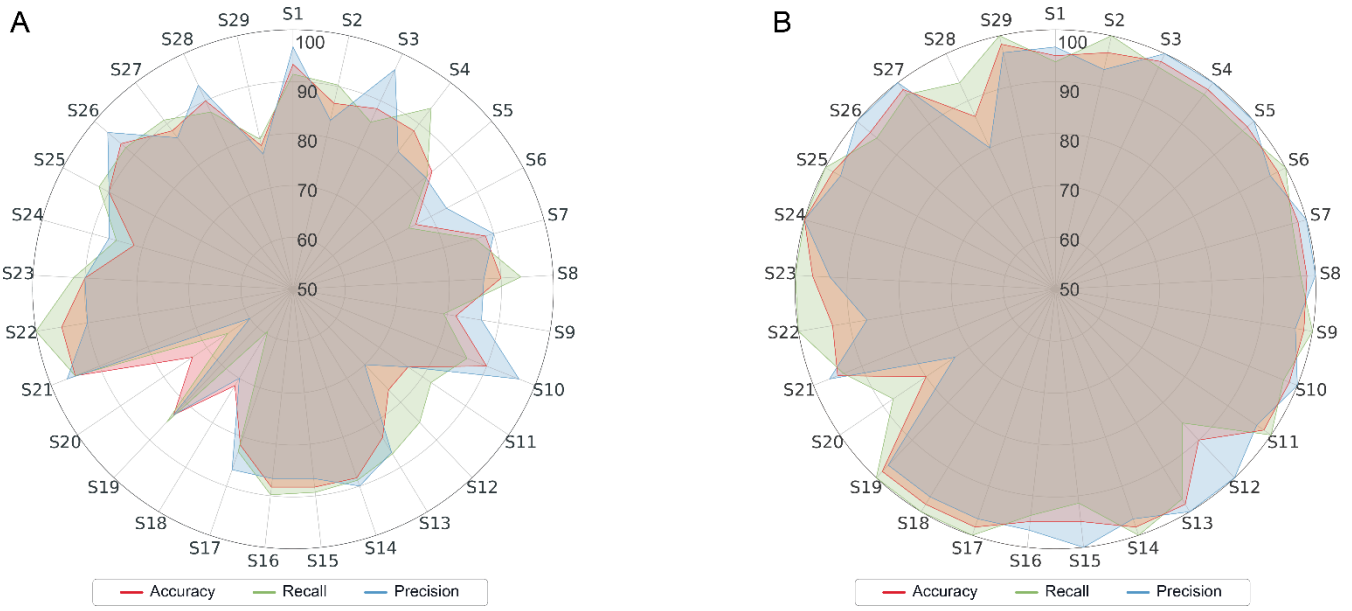


**Figure 1. Model performance trained on both datasets. (A) is about connectivity (CON) and (B) is about connectivity and PSD features (CON-PSD). Red, green, and blue donate accuracy, recall, and precision, respectively.**

repeatedly perform simple arithmetic calculations. Each subject underwent a total of 60 trials.

### *B. Preprocessing*

All data processing was conducted with Matlab R2020a (MathWorks, Natick, MA, USA) and BBCI toolbox [17]. First, the raw EEG signal was re-referenced using the common average reference. The recordings of each trial were segmented into 10-seconds epochs that contained a task period. The dataset was filtered by a band-pass of 1 to 40 Hz (fifth-order Butterworth filter). EOG channels were rejected for analysis.

### *C. EEG Connectivity and Power Spectral Density*

Edge and node features were extracted from EEG raw signals, because the GNN model receives graph-type data. For computing edge features, connectivity was utilized. Then, for extracting node features, the node strength and power spectral density were used. The connectivity was derived by the envelope correlation method to generate a connectivity matrix that represents the relationship of inter-node [18]. To compute envelope correlation, analytic signals were obtained by applying the Hilbert transform, and the mean of Pearson correlation between orthogonalizing values of two signals was calculated. The inter-node relations were evaluated via adjacency matrices that represent connectivity, distance, and degree information. In addition, the node strength was used as node feature. Power spectral density values of delta (1-4 Hz), theta (4-8 Hz), alpha (8-12 Hz), beta (12-30 Hz), and gamma (30-40Hz) frequency bands were utilized to derive node features. Ultimately, each graph yielded 435 edges above the main diagonal of the square matrix, and the dimension of node features was 30 * 6.

To compare the effect of PSD features on model performance, a dataset with connectivity (CON) only and a dataset with connectivity and PSD features (CON-PSD) were generated.

### *D. Experiments*

GNN model was trained by optimizing the negative log-likelihood loss function using Adam [19] and backpropagation. The 5-fold cross-validation method is followed to evaluate the performance of the trained model; the 5-folds average was calculated to evaluate the final model performance. To avoid possible overfitting problems, data augmentation was executed to increase the size of the training dataset size to ten times. Augmented data was generated by adding Gaussian noise with $\mu = 0$, and $\sigma = 0.001$ to the original data. As for the model evaluation, only the non-augmented original test dataset was used. Experiments were conducted once for each CON and CON-PSD features.

### *E. Classification using Graph Neural Networks*

To identify and classify the MA task-related brain activity patterns, this study utilized hierarchical graph pooling with a structure learning model (HGP-SL) [20]. HGP-SL is the state-of-the-art method for enhancing the effectiveness of GNN, and has shown notable performance on open datasets for graph classification (e.g., Mutagenicity [21], PROTEINS [22], etc.). HGP-SL enhances GNN by 1) graph pooling that effectively preserves a subset of informative nodes, the 2) structure learning that performs graph structure learning for the pooled subgraph.

Graph pooling operation by HGP-SL generates a subgraph by assigning an information score, which defined as:

$$p = \gamma(\varsigma_i) = \left\| \left( I_i - (D_i)^{-1} A_i \right) H_i \right\|_1 \qquad (1)$$

, where Ai and Hi were the adjacent and node representations matrices, and $\| \bullet \|1$ performs L1 norm row-wisely. Di denotes the diagonal degree matrix of Ai, and Ii the identity matrix. Subsequently, the pooled subgraph is generated by selecting a pooling ratio that of high-scored nodes. Edge and node features of the subgraph were imported as of selected nodes.

The other characteristic of HGP-SL, i.e., structure learning, derives edges by evaluating attention score. for nodes that disengage from each other after pooling, the graph structure information vanishes. To avoid this problem, the HGP-SL employs an attention mechanism that permutes the similarity between preserved nodes and can be formulated as follows:

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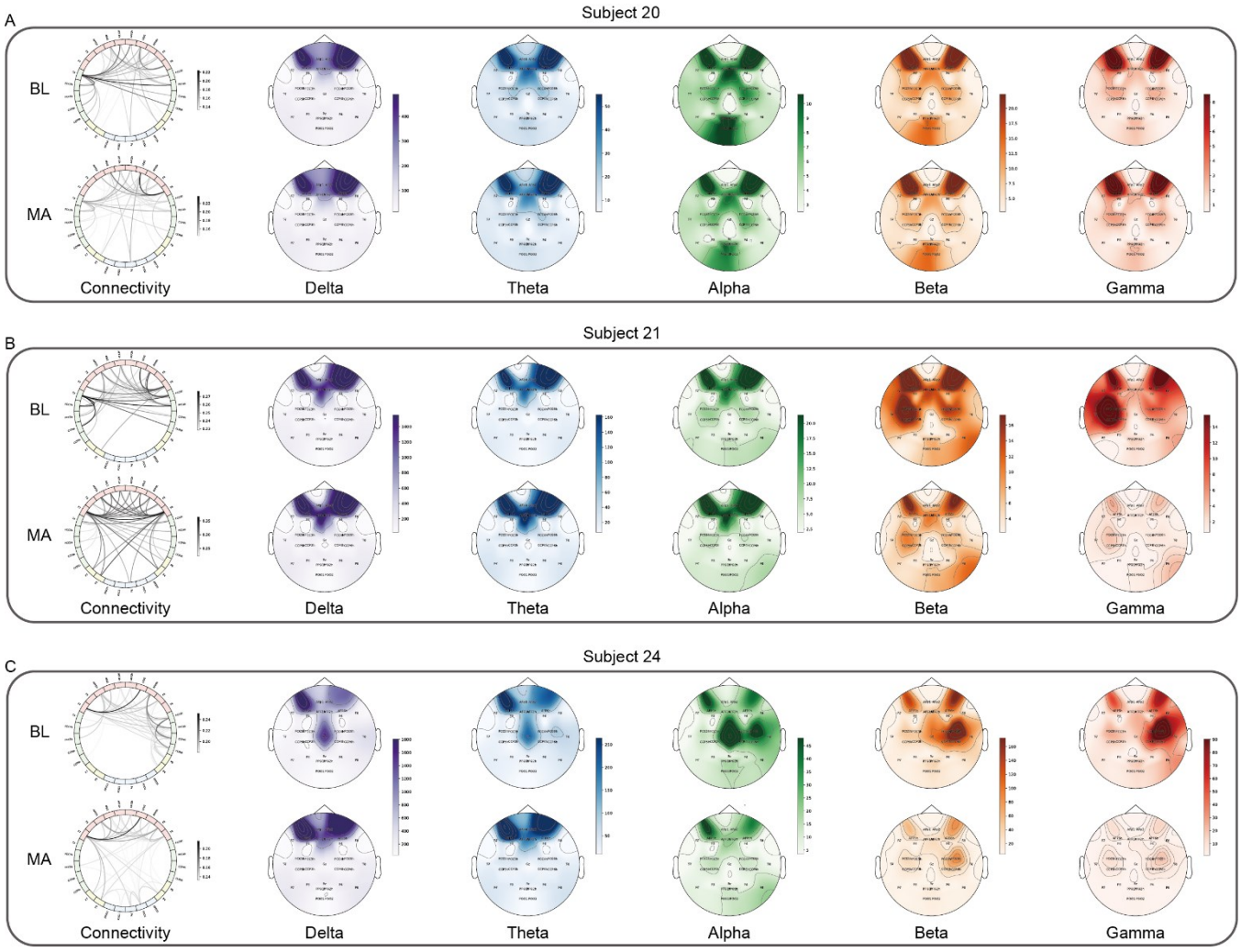


**Figure 2. Circular graphs depicting the connectivity of envelope correlation between nodes, and topographic visualization of each PSD band. (A), (B), and (C) were obtained from 20, 21, 24 subjects in CON-PSD dataset. The upper and lower rows represented each baseline (BL), mental arithmetic (MA). Delta, theta, alpha, beta, and gamma PSD bands were colored purple, blue, green, orange, and red, respectively.**

$$E_i(p,q) = \sigma\left(\vec{a}\left[H_i(p,:) \| H_i(q,:)\right]^{\bullet}\right) + \lambda \cdot A_i(p,q) \quad (2)$$

, where σ ( • ) is the activation function and || represents the concatenation operation. Hi(p, :) and Hi(q, :) indicate the p-th and q-th row of matrix HI, which denote the representations of node vp and vq, respectively.

Next, the similarity score was normalized to better compare the similarity between edges. Normalization was conducted via sparemax function that returns the Euclidean projection of input onto the probability simplex, and can be formulated as follows:

$$S_i(p,q) = sparsemax\left(E_i(p,q)\right) \quad (3)$$

$$sparsemax\left(E_i(p,q)\right) = \left[E_i(p,q) - \tau\left(E_i(p,:)\right)\right]_+ \quad (4)$$

, where [x]+ = max [23], and τ ( • ) was the threshold function that returned a threshold according to the procedure. Thus, sparsemax preserves the values above the threshold, and the other values will be truncated to zeros.

Graph level of hierarchical representation value was extracted from each layer by readout method that performed global mean pooling and global max pooling. The last layer (e.g., MLP and softmax layer) estimates the label using the concatenated value of representation after the HGP-SL layer.

### F. *GNNExplainer*

The GNNExplainer identified a subgraph structure that has a crucial role in trained GNN's classification [24]. The primary objective for the GNNExplainer is to generate a subgraph that explains the decision. This process is maximizing the mutual information between the subgraph and the computation graph. The secondary objective was the subgraph needs to be minimal. The GNNExplainer is literally the combination of prediction loss and edge size loss. The GNNExplainer was performed in the following order. In first, the computation graph which is the k-hops neighbor for node classification, or the entire graph for graph classification was extracted. Then, a mask for each edge in the computation graph was initialized. And then, a neural network that learns the mask with loss was constructed. Lastly, the mask was used, to reduce the computation graph to a minimal graph.

## III. Results

The HGP-SL enhanced GNN yielded better classification performance on CON-PSD compared to CON feature as shown in Figure 1. The mean accuracy was 85.57 ± 6.27 and 96.26 ± 4.14% for the CON feature and the CON-PSD feature, respectively. In CON feature, the best accuracy was found in subject 21 (95.00 ± 6.67%), whereas the lowest accuracy was

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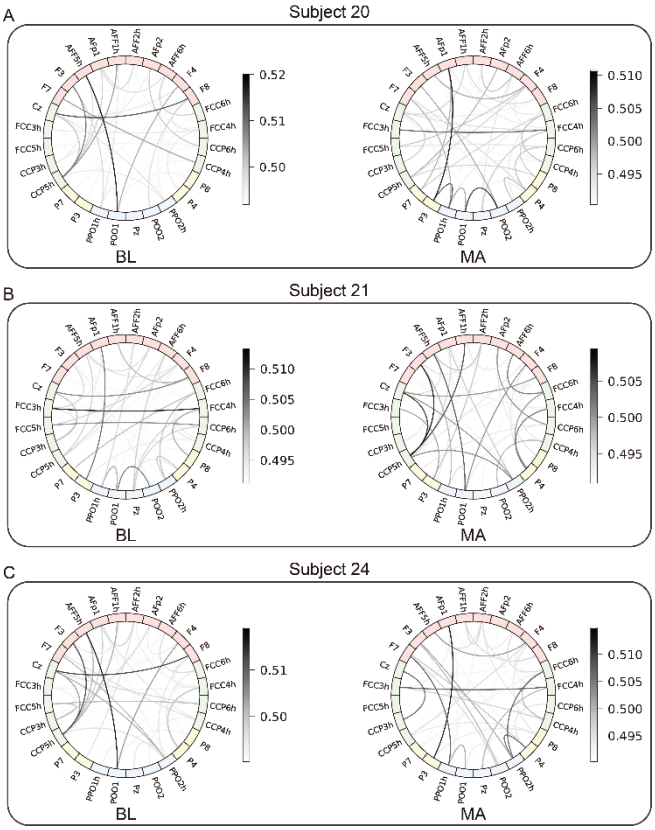


**Figure 3. Average of subgraphs that found as important in task classification by the GNNExplainer in (A) subject 20, (B) subject 21, and (C) subject 24.**

from subject 20 (73.33 ± 16.99%) presented in Fig. 1A. In CON-PSD feature, the best classification accuracy was found in subject 24 (100.00%), and The lowest performance was from subject 20 (80.00 ± 13.54%) illustrated in Fig. 1B.

During MA trials, certain trends were observed in connectivity and PSD. As for the connectivity, the connectome within frontal channels (AFp1, AFp2, AFF1h, AFF2h, AFF5h, AFF6h, F3, F4, F7, and F8) was increased or maintained, as illustrated in Figure 2. In addition, the connection within the parieto-occipital lesion (P4, P8, PPO2h, POO2, Pz, POO1, PPO1h, P3, and P7) was stronger than baseline. In terms of PSD, theta-band power at the frontal electrode had increased, whereas beta-band power had decreased in the central region.

Certain subgraph patterns that derived by the GNNExplainer, such as Figure 3.A, C in BL, and Fig 3.B in MA, were observed in all subjects. Two prominent patterns were 1) from CCP5h to the frontal regions (AFF1h, AFp1, AFF5h, and F3), and 2) from AFF5h to POO1.

## IV. Discussion and Conclusion

Adding PSD to CON yielded a significant increase in classification performance; even for subject 20, where both models returned the lowest accuracy, CON-PSD still yielded a slightly better accuracy (73.33 ± 16.99 vs. 80.00 ± 13.54 %). The results suggest that by considering not only the connectivity but also the power density of the frequency bands, higher performance can be achieved for MA classification using the GNN model.

Previous studies regarding the connectivity during MA tasks reported that the connectivity of the frontal and the left parietal regions are associated with cognitive workload, and active involvement of left parietal area during MA [25-27]. The findings of this study are consistent with previous studies; the connection within the frontal region was sustained or increased, and the parieto-occipital lesion (P4, P8, PPO2h, POO2, Pz, POO1, PPO1h, P3, and P7) was included in the connectivity. Further, there was an incremental tendency in the theta band power of the frontal region, similar to a previous MA study which suggested that high frontal theta activities are related to cognitive control and concentration [25]. In addition, there was a tendency that the beta band power in the central cortex decreased, which is suspected to associated with mental workload [25, 28].

Among the connections between nodes identified as important by GNNExplainer, two notable patterns were found in Fig. 3: edges from CCP5h to frontal regions (AFF1h, AFp1, AFF5h, and F3), and the frontoparietal connection from AFF5h to POO1. The former could be reflecting left superior parietal gyrus activity during performing subtraction [29], whereas the latter reflecting frontoparietal connectivity which is associated with effective mental arithmetic tasks [30]. The findings of this study indicate that the connectivity patterns reported here could be important in MA classification, and that the connectivity analysis by “explainable” GNN could be beneficial in investigating brain activity patterns during cognitively demanding tasks.

Limitations of this study should be mentioned. First, this research was performed in an offline environment with retrospective data. Second, the adding PSD information resulted in significant increase in classification performance, however the explainability measure utilized in this study was not applicable to PSD features. Future studies with similar design should aim to validate the proposed methods, preferably with a real-time, prospective design in a larger cohort.

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